



9. Design and implement C/C++ Program to sort a given set of n integer elements using Selection Sort method and compute its time complexity. Run the program for varied values of $n > 5000$ and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator. #include <stdio.h>

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
// Function to perform selection sort
```

```
void selectionSort(int arr[], int n) {
```

```
int i, j, minIndex, temp;
```

```
for (i = 0; i < n - 1; i++) {
```

```
minIndex = i;
```

```
for (j = i + 1; j < n; j++) {
```

```
if (arr[j] < arr[minIndex]) {
```

```
minIndex = j;
```

```
}
```

```
}
```

```
// Swap the found minimum element with the first element
```

```
temp = arr[minIndex];
```

```
arr[minIndex] = arr[i];
```

```
arr[i] = temp;
```

```
}
```

```
}
```

```
// Function to generate random numbers between 0 and 999
```

```
int generateRandomNumber() {
```

```
return rand() % 1000;
```

```
}
```




```
int main() {

// Set n value

int n = 6000;

// Allocate memory for the array

int* arr = (int*)malloc(n * sizeof(int));

// Generate random elements for the array

srand(time(NULL));

printf("Random numbers for n = %d:\n", n);

for (int i = 0; i < n; i++) {

arr[i] = generateRandomNumber();

printf("%d ", arr[i]);

}

printf("\n");

// Record the start time

clock_t start = clock();

// Perform selection sort

selectionSort(arr, n);

// Record the end time

clock_t end = clock();

// Calculate the time taken for sorting

double time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

// Output the time taken to sort for the current value of n

printf("\nTime taken to sort for n = %d: %lf seconds\n\n", n, time_taken);

// Display sorted numbers

printf("Sorted numbers for n = %d:\n", n);

for (int i = 0; i < n; i++) {
```




GSSS

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```
printf("%d ", arr[i]);
```

```
}
```

```
printf("\n\n");
```

```
// Free the dynamically allocated memory
```

```
free(arr);
```

```
return 0;
```

```
}
```